

1 Site-level rates and site-level coefficients estimate different
2 quantities: a record-level benchmark for federated analysis
3 of COVID-19 surveillance data

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5 **Abstract**

6 **Objective:** Federated networks share either site-level rates or coefficients from site-level models,
7 and treat these as interchangeable. A regression across site rates absorbs every between-site
8 difference that tracks the regressor, so it need not recover the within-site association a coefficient
9 estimates. We tested whether the two agree.

10 **Materials and Methods:** We analyzed 3 993 464 confirmed COVID-19 case records from Mex-
11 ico's surveillance system, 32 treating-unit states. The observed ecological slope regressed state
12 case fatality on the share aged 70 or older. The model-implied slope averaged record-level lo-
13 gistic predictions within each state and regressed those on the same share. Their difference is
14 the ecological discrepancy. We varied the record-level model across six prespecified specifica-
15 tions and the site definition across residence state, sector, municipality, and a random partition
16 holding each state's size and elderly count fixed.

Results: Under the prespecified three-covariate model the observed and model-implied slopes were 1.31 and 0.45 percentage points of case fatality per point of elderly share (difference 0.86; bootstrap 95% CI 0.50–1.30). The difference fell as adjustment increased but remained 0.59 (0.21–1.07) under the richest specification. On matched records it was 0.47 across municipalities against 1.01 across their states of residence, and 0.00 under the random partition, by construction. Aligned one-stage and two-stage estimators agreed to 0.0007 in the log odds ratio.

Discussion: The discrepancy is defined relative to a specified record-level model and does not identify a contextual effect.

Conclusion: Networks sharing rates should name the record-level model their inference assumes, and share coefficients where a record-level association is the target.

Keywords: ecological fallacy; multicenter studies as topic; data aggregation; electronic health records; COVID-19

1 Background and Significance

Federated networks are central infrastructure for multi-site clinical research: ENACT/ACT across the CTSA consortium [1], OHDSI across a common data model [2], TriNetX across more than 20 countries [3], and PCORnet across partner networks serving over 47 million patients [4]. These networks share different objects, and the ecological fallacy [5] arises in the subset of arrangements where a relationship among site rates is read as an association among patients: rate and prevalence dashboards, consortia that pool aggregate counts, and any analysis regressing site rates on site composition. Networks that fit local models and pool coefficients are not exposed in the same way, which Table 1 bears out. The mechanisms of the fallacy are well understood [6, 7] and formal frameworks for ecological inference exist [8–10], but how far a working federated analysis departs from the record-level association it is read as estimating has rarely been measured on data where both can be computed. Between-site differences between observed and predicted outcomes are studied as calibration in multicenter prediction research [11, 12]. The quantity measured here is a projection of those differences onto one site-level covariate, on the sites the record-level model was fitted on rather than on new centers.

What a network shares is not one choice but several, and they do not preserve the same thing. A network may share site-level rates and prevalences, or coefficients from a model each site fits locally.

These are routinely treated as interchangeable summaries of the same underlying association. They are not: a regression across site rates estimates a relationship between site-level quantities, while a coefficient estimates a within-site association among records, and the two coincide only under conditions the network cannot check from those marginal summaries alone.

The machinery for relating the two has existed for some time. Aggregate rates can be derived from an individual-level disease model by integrating it over each area’s covariate distribution, which is the basis of aggregate-data and hierarchical related regression [13–15]. Reanalyses of Robinson’s own 1930 census data find state variation in illiteracy that the states’ race and nativity composition does not account for [16], and finer areal units are known to reduce the discrepancy [17, 18]. What has not been reported is the comparison run on a complete record set where both quantities are computable and the shared object is varied deliberately: the slope observed across site rates, against the slope the fitted record-level model implies for the identical sites and records, with the site definition and the record-level model each varied over a prespecified range.

We report that comparison on 3 993 464 confirmed COVID-19 case records from Mexico, and make three claims. (1) The observed slope across treating-unit states exceeds the slope implied by the registered record-level model for the same states by 0.86 percentage points of case fatality per point of elderly share, and the difference remains positive under every richer specification we registered. (2) The difference is indexed to that model rather than being a property of the data alone: it falls from 0.98 to 0.59 across six prespecified specifications, and a model carrying a free term per state would remove it entirely, so the quantity measures state-structured lack of fit. (3) One-stage and two-stage estimators of the record-level association agree to 0.0007 in the log odds ratio when their specifications are aligned, so what moves the estimate is which object the network shares, not which estimator it uses.

2 Materials and Methods

2.1 Reporting guidelines

This study uses routinely collected health data. We follow the RECORD extension of the STROBE statement for studies conducted using observational routinely collected health data [19]. The unit of analysis is a case record: the source carries a record identifier and no person identifier, so repeat

presentations by one patient cannot be collapsed and are not treated as such.

2.2 Data and provenance

Mexico’s Secretaría de Salud publishes individual-level COVID-19 surveillance data covering all confirmed and suspected cases nationwide. We used the January 3, 2022 historical snapshot (12 425 179 records, 3 993 464 confirmed cases, 299 581 deaths), pinned by SHA-256 and verified on every run. Each record carries age, sex, state of the treating medical unit, state and municipality of residence, the delivering sector, symptom-onset date, death date and nine comorbidity flags. Cases were classified as confirmed when `CLASIFICACION_FINAL` $\in \{1, 2, 3\}$. The 32 treating-unit states range from 22 307 records (Chiapas) to 1 154 276 (Mexico City). The loader enforces the published catalogue’s validity domain and reports every code outside it; 2874 records carry no municipality and 20 no sector, and these are excluded only from the site definitions that need them.

Analyses ran in a container that verifies, against a manifest the launcher supplies, that the analysis files it holds are the committed ones before any of them executes. Each reported number is bound to the run that produced it in a hash-chained ledger, which is in the repository.

2.3 Three architectures on one cohort

A federated network never holds the records, so it cannot compare what its shared summaries preserve against what the records support. We reproduce the three architectures on one complete record set, which allows both sides of the comparison to be computed on identical records. *Rate sharing*: each site contributes its case fatality and its share of records aged 70 or older, and a regression across sites estimates the relationship between them. *Coefficient sharing*: each site fits the same logistic model locally and contributes its coefficients and their standard errors, which are pooled by inverse-variance meta-analysis. *Centralized*: the records are pooled and one model is fitted. The second and third are compared under matched specifications, since an unaligned pair differs by its target rather than by its architecture.

2.4 Statistical analyses

The two slopes. The observed ecological slope is an unweighted ordinary least-squares regression of site case fatality on the site share of records aged 70 or older. Unweighted fitting is the

registered primary specification; its estimand is the relationship across sites treated as equal units, and binomial and record-count-weighted fits are reported beside it in the supplement, where the record-count-weighted discrepancy across treating-unit states is +1.3215 against the equal-state +0.8568. The model-implied slope fits the record-level logistic model, averages its predicted probabilities within each site, and regresses those averages on the same share over the same sites. The difference between the two slopes is the ecological discrepancy. Its interval is a bootstrap over sites with the record-level model refitted inside every draw; the ordinary least-squares standard error of the implied slope is not an uncertainty for it, because its response is fitted rather than observed.

Record-level models. Six specifications of increasing adjustment, registered before any was fitted: age 70 or older alone; plus sex and any of nine comorbidities; the nine entered separately; age in ten-year bands; age as a restricted cubic spline with five knots at the 5th, 27.5th, 50th, 72.5th and 95th percentiles; and that model plus calendar month of symptom onset. Estimation used `statsmodels` 0.14.6.

Site definitions. Treating-unit state, state of residence, health-care sector, and municipality of residence at four registered size thresholds, each paired against the states those municipalities nest in on the same records, with the bootstrap clustered on state of residence. A matched random partition draws pseudo-sites that reproduce every real state’s size and count aged 70 or older exactly, using no other record field, over 500 replications.

Pooling. Per-site logistic coefficients were pooled by inverse-variance meta-analysis under common-effect, DerSimonian–Laird and REML with Hartung–Knapp specifications, and compared against one-stage fits with state fixed effects, a random intercept and a random slope. Inverse-variance meta-analysis of separately fitted coefficients targets a precision-weighted summary and is not distributed score aggregation, so the agreement reported here is an empirical result on these data rather than a preservation property.

The state-level model. A binomial random-intercept model over cells of state, age, sex and the nine comorbidities, carrying the state’s standardized elderly share alongside the record-level covariates, fitted by `lme4::glmer` 1.1-31 under R 4.2.2 at 15 adaptive Gauss–Hermite nodes, and refitted by a second implementation of the same quadrature written for this paper. Eleven numerical acceptance criteria, fixed before the fit, govern whether the result is reported as evaluable.

2.5 Registration and multiplicity

Ten supplementary analyses were registered in a protocol frozen at commit 97f7094 before any of them was run, each tagged by whether its outcome was already known, with falsification criteria and the main-text and supplement split fixed in advance. The site definitions and the covariate ladder were registered in a second protocol frozen at commit a38196b, which records the state-level discrepancy already known at that point and states that no grouping other than the treating unit's state had been run. Both protocols, the implementation amendments and a correction record are in the repository. All registered analyses are reported regardless of outcome.

2.6 Reproducibility

All code, the analysis scripts, the frozen protocols and the ledger binding each reported number to its run are archived on Zenodo (DOI: 10.5281/zenodo.21252372) and available at <https://github.com/elliotttower/ecological-bias-covid>. The Mexico data is published by the Dirección General de Epidemiología at datos.gob.mx; the supplementary United States data is at data.cdc.gov, dataset vbim-akqf.

3 Results

3.1 Rate sharing against coefficient sharing

Across the 32 treating-unit states, state case fatality rises with the share of records aged 70 or older at +1.3105 percentage points per point of elderly share (SE 0.2373, $R^2 = 0.50$). Averaging the registered record-level model's predicted probabilities within each state and regressing those averages on the same share gives +0.4537. Both are unweighted linear projections of a state rate on state elderly share over the same 32 states (Figure 1), so the difference between them, +0.8568 (bootstrap 95% CI over states 0.4956 to 1.3027), is the ecological discrepancy under the registered model. The registered falsification criterion, that the two agree within 20%—a margin fixed before the analysis and wide enough that a discrepancy inside it would not change how a network should read its summaries—is not met.

Sharing coefficients instead reproduces the record-level association (Table 1). Estimators sep-

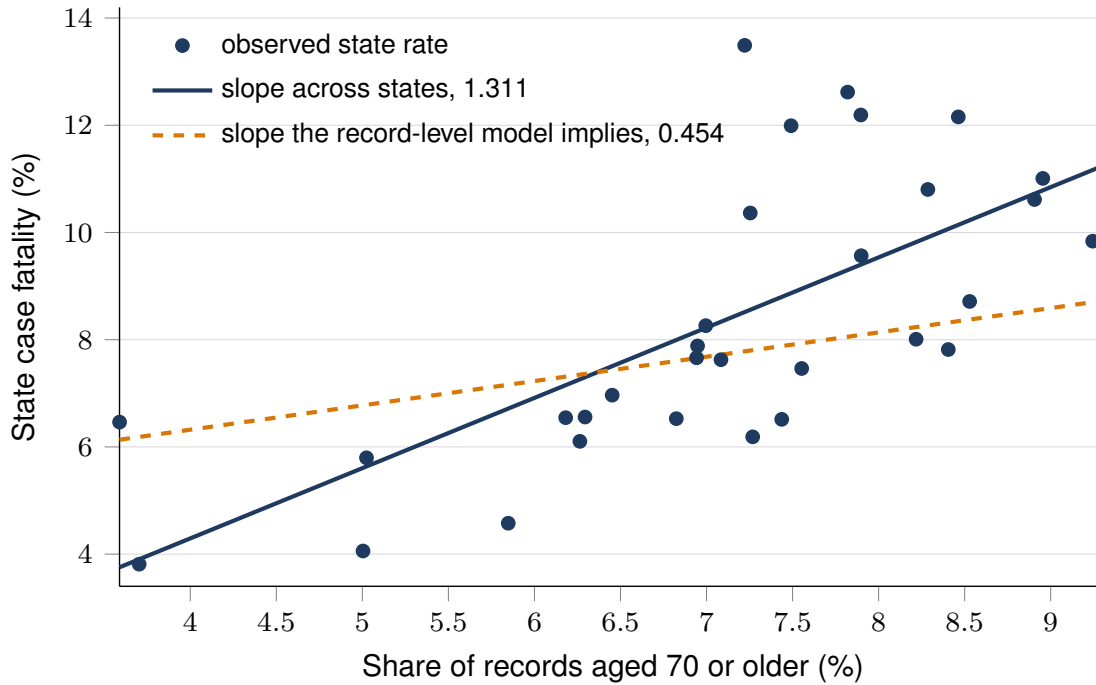


Figure 1: Each point is one of Mexico’s 32 treating-unit states, 3 993 464 confirmed COVID-19 case records. Points are each state’s observed case fatality. The solid line is their regression on elderly share. The dashed line is the regression, over the same 32 states, of the rate implied by averaging a record-level logistic model’s predicted probabilities over that state’s own patients, under the registered model carrying age 70 or older, sex and any of nine comorbidities; the implied rates themselves are not plotted. The two slopes are therefore comparable: 1.311 observed against 0.454 implied. Their difference, 0.857 percentage points of case fatality per point of elderly share, is the ecological discrepancy.

Alt text: Scatter plot of 32 Mexican states, with the share of records aged 70 or older on the horizontal axis and state case fatality on the vertical axis. Points are each state’s observed case fatality. A solid line through them rises steeply, with slope 1.311. A dashed line, the slope the record-level model implies for the same states, is much flatter at 0.454. The two lines cross near the middle of the range and diverge toward both ends, so the relationship observed across states is far steeper than the one the record-level model implies.

156 arate by the quantity they target rather than by whether the fit is one-stage or two-stage: four
157 common-effect specifications fall between odds ratios 11.26 and 11.28 and three random-effects
158 specifications between 11.73 and 11.74, while the two aligned pairs differ by 0.00015 and 0.00068 in
159 the log odds ratio. Heterogeneity is preserved rather than hidden: per-state odds ratios run from
160 7.99 to 18.09, $I^2 = 97.4\%$, with a prediction interval of 8.43 to 16.32.

Table 1: Estimators of the record-level association between age 70 or older and death, Mexico, 3 993 464 records in 32 treating-unit states. Specifications are grouped by the quantity they target. Within each group, the one-stage and two-stage fits differ by less than 0.0007 in the log odds ratio; between groups they differ by 4.2%, which is the common-effect against random-effects choice and not the one-stage against two-stage one.

Target	Specification	Odds ratio
<i>Common effect</i>		
	Pooled record level	11.264
	One-stage, state fixed effects	11.275
	One-stage, random intercept	11.275
	Two-stage, common effect	11.276
<i>Random effects</i>		
	Two-stage, DerSimonian–Laird	11.729
	Two-stage, REML, Hartung–Knapp	11.737
	One-stage, random slope	11.737

3.2 Dependence on the record-level model

The discrepancy is defined against a model, and it shrinks as that model is enriched (Table 2). Entering age as a spline with the nine comorbidities separately moves the implied slope from +0.4537 to +0.6549 and the discrepancy from +0.8568 to +0.6557; adding calendar month of onset leaves +0.5941. The bootstrap interval excludes zero at every rung. A model carrying a free term per state would reproduce each state’s rate exactly and remove the discrepancy altogether, so the quantity measures state-structured lack of fit of a specified model rather than a model-free amount of ecological bias.

3.3 Dependence on the site definition

The discrepancy is also a property of how sites are drawn (Table 3). It is +0.8497 across the 32 states of residence and +0.7657 across the 11 health-care sectors. On the 3 665 523 records retained by the municipality size rule, it is +0.4689 across 385 municipalities against +1.0087 across the states those municipalities nest in, a paired difference of -0.5398 (state-clustered bootstrap -1.1642 to -0.1299). The discrepancy is smaller at finer grain, consistent with what is reported for nested residential geographies [17, 18], although elderly share is measured from the same records and is

Table 2: Observed minus model-implied ecological slope across six prespecified record-level models, in percentage points of case fatality per point of elderly share. The observed slope is +1.3105 throughout. Intervals are a state bootstrap with the record-level model refitted in each draw. Specifications were registered at commit [a38196b](#) before any was fitted.

Record-level model	Implied	Difference	95% CI
Age 70+	0.332	0.979	0.605–1.458
+ sex, any comorbidity (<i>registered</i>)	0.454	0.857	0.496–1.303
+ nine comorbidities separately	0.495	0.816	0.477–1.243
age in ten-year bands instead	0.650	0.661	0.269–1.148
age as a restricted cubic spline instead	0.655	0.656	0.260–1.149
+ calendar month of onset	0.716	0.594	0.212–1.070

Table 3: Observed and model-implied ecological slopes under each prespecified site definition, with the registered three-covariate record-level model. The municipality row is restricted to municipalities with at least 1000 records and is paired against the states of residence those municipalities nest in, on the same records.

Site definition	Sites	Observed	Implied	Difference
Treating-unit state	32	1.311	0.454	0.857
State of residence	32	1.305	0.455	0.850
Health-care sector	11	1.193	0.428	0.766
Municipality (≥ 1000 records)	385	—	—	0.469
paired states of residence	32	—	—	1.009

estimated less precisely in small units.

A registered reassignment of records to pseudo-sites holding each state’s size and elderly count fixed returns a median discrepancy of +0.0005 over 500 replications. Its expectation is zero by the estimating equations, so it checks the construction rather than geography, and the supplement reports it in full.

3.4 A residual state-level association

Fitting the state’s standardized elderly share alongside the record-level covariates of the richest composition model, as a binomial random-intercept model over 152919 cells by adaptive Gauss–Hermite quadrature at 15 nodes, leaves an odds ratio of 1.1817 per standard deviation (profile-likelihood interval 1.0719 to 1.3027) with a random-intercept standard deviation of 0.2682. The fit

meets all eleven numerical acceptance criteria fixed before it was run, and a second implementation of the same quadrature, written separately and run on the same cells, reproduces its coefficients to 7.7×10^{-5} . The association is estimated from 32 clusters and conditions on the measured record-level covariates only, so it is a model-based residual association and not an identified contextual effect.

4 Discussion

4.1 What the difference contains

A regression across site rates absorbs any between-site variation in the outcome that correlates with the regressor, whatever produces it. The slope implied by a record-level model includes only what that model attributes to the covariates it holds, averaged over each site’s own covariate distribution. Their difference is therefore the component of what the site rates contain, aligned with elderly share, that the record-level model does not reproduce, and the covariate ladder bounds it from one side, showing that measured composition accounts for part of it and leaves the rest.

What remains is not attributable from this design. It contains unmeasured composition, differences in case ascertainment and coding between states, the point each state had reached in its epidemic, any genuine effect of where a patient was treated, and misspecification of the record-level model, which enters the residual by construction. Noncollapsibility of the odds ratio [20] does not contaminate the comparison, because the implied slope is formed from predicted probabilities rather than from a conditional coefficient; it does bear on any attempt to read the record-level odds ratio as the quantity the ecological slope is estimating.

Nothing in the comparison is specific to COVID-19. COVID-19 serves here because the available record-level data permit both halves of the comparison on the same cohort; the same structure applies to drug-safety surveillance, federated genomic association, and any multi-site study pooling site-level summaries.

4.2 Implications for federated analysis infrastructure

A network that shares site rates and a network that shares per-site coefficients are not doing the same analysis at different levels of privacy. The rate regression estimates a relationship between

site-level quantities; the coefficient estimates a within-site association among records. On these data the second is recoverable from shared summaries to within 0.0007 in the log odds ratio, and the first includes an additional +0.8568 that no record-level model we registered reproduces. Which object a network shares therefore determines what its central analysis can estimate, and the choice is an estimand question before it is a privacy or engineering one.

That does not make rate sharing wrong. A cross-site rate relationship is a well-defined quantity and may be the one a surveillance question asks for. It is wrong only when read as the association among patients, and a network cannot tell the two apart from site rates alone: the discrepancy is invisible from those marginal summaries. Four routes avoid the substitution.

Joint stratum counts. Every covariate in the specifications reported here takes finitely many values, so the record-level fit, the model-implied slope and the discrepancy are computable from counts of deaths and records in each joint stratum of age, sex, comorbidity and month within each site; the state-level model is fitted over 152 919 such cells. A coordinating center that receives joint counts rather than marginal rates can compute the benchmark without individual records. Joint counts remain potentially disclosive because a fine stratum may contain one person; minimum cell sizes, suppression, or other network-appropriate disclosure controls are still required.

Federated regression. Fed-GLMM [21] and DataSHIELD [22] enable record-level estimation while sharing only summary statistics: sites run a standardized script locally and share coefficients, supporting record-level estimation without the aggregation substitution studied here.

Large individual-level cohorts. N3C [23] holds individual records for direct estimation across more than 75 sites.

Hierarchical models. Multilevel models with site random effects represent both levels explicitly rather than choosing one [24], and shrinkage toward the group mean limits the inflation of extreme site-level estimates [25].

4.3 Practical guidelines for multi-site studies

Name the estimand the shared object supports. A protocol that says a network will “pool site-level data” has not said what it will estimate. Site rates support a cross-site relationship; per-site coefficients support a within-site association. Writing which one the study targets, before the summaries are collected, is the step that makes the rest checkable.

Report heterogeneity rather than averaging it away. A pooled odds ratio of 11.73 is the right answer to the question it answers; the prediction interval of 8.43 to 16.32 is what tells a reader whether that is the question they have.

Share coefficients when the target is a record-level association. The agreement in Table 1 holds for compatible local models, harmonized coding and adequate cell counts at each site. A network has to verify those rather than assume them.

Register the benchmark, not only the procedure. Where a reported quantity is a comparison against a model, the model is part of what is being reported. The discrepancy here moves from +0.979 to +0.594 across six record-level specifications. Weighting moves it further, from +0.857 under equal-state fitting to +1.322 under record-count fitting, and that movement reflects a change in the target relationship, not instability of a single estimate. A registration that fixes the procedure and leaves the benchmark and the weighting open has not fixed the number.

5 Limitations

No specification is established as correct. The discrepancy is the part of the cross-site rate relationship a given record-level model does not reproduce, so a better model would leave less of it and a model with a free term per state would leave none. The paper reports the range across six registered specifications and does not identify a true one.

A discrepancy near zero does not certify the site predictions. The statistic is the association between elderly share and the aggregated residual, so it captures only the component of site-level error aligned with that share. Three equally weighted sites with equally spaced elderly shares and residuals of +5, −10 and +5 percentage points give a residual slope of exactly zero: the equal endpoint errors cancel in the slope, while the middle error falls at the mean elderly share.

The residual is not attributable. This design cannot separate the sources enumerated in Section 4.

The diagnostic is not evaluated as a guide to action. Whether acting on a nonzero discrepancy improves later predictions is a separate question from whether the discrepancy is present,

and these analyses do not test that use. Answering it requires temporally separated evaluation against ordinary recalibration.

Finer units are measured less precisely. Elderly share is estimated from the same records, so the municipality comparison confounds geographic grain with measurement error in the regressor. The grouped-binomial fit reported beside the equal-state one addresses outcome variance in small municipalities, not error in the regressor, which nothing here corrects.

The architectures are reproduced, not observed. A live network differs in local model specification, coding harmonization and per-site cell counts, and the agreement between pooling strategies reported here is conditional on those being adequate. The source carries record identifiers and no person identifiers, so repeat presentations by one patient are counted separately.

6 Conclusion

A network that shares site rates cannot find this discrepancy in what it shares. The observed slope and its interval are computable from the rates; the benchmark they would have to be set against is not, because forming it requires the joint distribution of the covariates within each site. Nothing in the shared rates indicates whether the gap is 0.86 or zero, and the joint counts that would carry it are a different summary rather than the records themselves.

The choice of shared object therefore has to be made when a network is designed, from the estimand the study will report, and cannot be audited afterward from the summaries it produced. Where the target is an association among patients, per-site coefficients from aligned local models carry it; where the target is a relationship between sites, rates carry that, and the two should not be read as estimates of each other.

7 Supplementary Materials

Available online: (A) the two frozen protocols, the implementation amendments and the correction record, with a provenance table giving for each analysis its status, what was known when it was specified, and the commit that froze it; (B) an accounting table reconciling every denominator the paper uses, and a table of superseded values with the cause of each; (C) the ten registered supplementary analyses with their falsification criteria and outcomes, reported regardless of result;

(D) per-state estimates, the binomial and record-count-weighted sensitivities beside the equal-state primary fit, and the municipality analysis at all four registered size thresholds; (E) the full numerical diagnostics for the state-level model, including the second implementation and the eleven acceptance criteria; (F) two further record sets analyzed under the same design, United States case surveillance grouped by race and ethnicity and the 4CE consortium’s aggregate neurological data, which are not part of the analysis reported here.

Declarations

Ethics approval and consent to participate

This study used only publicly available, de-identified datasets. No ethics approval or informed consent was required. The CDC Case Surveillance Public Use Dataset is released under a public domain license. The Mexico Datos Abiertos dataset is released by the Secretaría de Salud for public use. The United States and 4CE data appear only in the supplement; the 4CE material is the aggregate site-level summaries published in [26].

Consent for publication

Not applicable.

Availability of data and materials

The Mexico COVID-19 surveillance data is available from `datos.gob.mx` [27], and the CDC COVID-19 Case Surveillance Public Use Dataset from `data.cdc.gov` [28]. The 4CE aggregate data is available in the supplementary materials of [26]. Neither the CDC nor the 4CE records are part of the analysis reported here; both are described in the supplement. All analysis code, the frozen pre-registration protocol, and computed results are available at <https://github.com/elliotttower/ecological-bias-covid>. A permanent archive is deposited at Zenodo (DOI: 10.5281/zenodo.21252372).

Competing interests

The author declares no competing interests.

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Authors' contributions

ET conceived the study, designed the analyses, wrote the code, performed all statistical analyses, and wrote the manuscript. The author read and approved the final manuscript.

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The research question, the two pre-registration protocols, the choice of specifications and site definitions, and all interpretations are the author's. Claude Code (Anthropic, versions 2.1.220 to 2.1.287, running the models Claude Opus 4.6, Claude Opus 5 and Claude Opus 5.5) was used between July and October 2026 to draft and edit manuscript text and to write and debug the analysis code. Perplexity AI (web application) was used for literature search and to review near-final drafts. The author directed and reviewed all of that work, verified every reported number against the record that produced it, and takes full responsibility for the accuracy and integrity of the work. No AI tool is listed as an author, and no AI-generated content was used as a primary data source.

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